

Shubhojyoti Datta Chaudhuri

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EXPERIENCE

LookThePart, Remote

Software Developer | Jan 2026 - Jun 2026

- Shipped 10+ new RESTful API endpoints and maintained 50+ existing endpoints in Python with Django REST Framework across authentication, product search, post sharing, and media handling, serving web and mobile clients from a single backend service.
- Modeled 15+ relational database entities in PostgreSQL using the Django ORM across entertainment, character, product, post, and media domains to support scalable content workflows.
- Engineered Pillow-based image processing and thumbnail generation into an existing client-owned AWS S3 media pipeline, handling upload and optimized cloud delivery for a store of 10,000+ image assets.
- Owned production defects end to end, diagnosing and resolving 504 gateway timeouts on long-running publishing operations plus 2 ordering defects in null-priority and exact-match sorting, improving reliability of editorial workflows.
- Collaborated with a cross-functional team across a 6-month agile software development lifecycle, contributing Git-based code reviews, iterative design discussions, and technical documentation.

VertexPlus Technologies Limited, India

Software Engineer (Intern) | May 2024 - Jun 2025

- Developed Python backend components for an automotive data-processing platform in collaboration with a cross-functional engineering team, enhancing data processing and application workflows.
- Integrated TensorFlow-powered OCR workflows into a Django-based web interface, automating document processing across the platform.
- Enhanced OCR and image-processing pipelines with custom character purification and page orientation detection and correction, improving text recognition for rotated and misaligned document images.
- Built a Python vehicle intelligence scoring algorithm and integrated reusable, object-oriented scoring logic into the application workflow to automate vehicle assessment.

PROJECTS

PintOS Thread Scheduler and User Programs | C, PintOS Kernel, GCC, QEMU, Linux

- Extended the PintOS operating system kernel in C with priority scheduling and priority donation, modifying thread state and 3 synchronization primitives (locks, semaphores, condition variables) to preserve correctness under lock contention.
- Implemented and debugged 13 system call handlers spanning process execution, argument passing, file descriptor management, and user memory validation to isolate user programs from kernel memory.

Cashback Reconciler - Rewards Reconciliation Engine | Python, PyTest, Plaid, LLMs

- Designed a deterministic matching algorithm that normalizes merchant names and reconciles card-linked offers against posted transactions, classifying rewards across 4 states: posted, pending, missing, or ineligible.
- Wrote clean, maintainable, object-oriented Python validated by 16 PyTest unit tests covering merchant normalization, transaction matching, and issuer reliability scoring, and scaffolded LLM-based email extraction with Plaid integration.

Sentiment and Emotion Analyzer - AI Model Evaluation Platform | React, Flask, Hugging Face Transformers, PyTorch

- Delivered a full-stack platform with a Flask REST API and React front end that evaluates and compares predictions from 5 transformer models on user-submitted text.
- Generated ensemble classifications across 5 transformer models through majority voting with confidence scoring and disagreement detection to improve evaluation reliability.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, JavaScript, SQL, HTML, CSS

Computer Science Fundamentals: Data Structures, Algorithms, Object-Oriented Design, Design Patterns, Operating Systems, Concurrency, Multithreading, Distributed Systems

Backend and Distributed Systems: Django, Django REST Framework, Flask, REST APIs, Django ORM, React, Apache Spark, PySpark, Hadoop, HDFS

Cloud and Databases: AWS S3, Docker, Linux, MySQL, PostgreSQL, SQL Databases, Firebase, NoSQL Databases

AI and Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, LLMs (Google Gemini), OCR, Model Evaluation, Machine Learning

Engineering Practices: Agile, Software Development Lifecycle (SDLC), Git, Version Control, Code Review, Unit Testing (PyTest), Debugging, Troubleshooting, Technical Documentation, Postman

EDUCATION

University at Buffalo, The State University of New York, Buffalo, NY

Master of Science in Computer Science, AI/ML Track | GPA: 3.7/4.0 | Aug 2025 - Dec 2026

Chandigarh University, India

Bachelor of Engineering in Computer Science | GPA: 3.2/4.0 | Aug 2021 - Jul 2025

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Distributed Systems, Database Systems, Machine Learning, Distributed Data Management